

References & Prior Work — AI-Native Self-Healing O-RAN using a Digital Twin

Digital-Twin-Verified Self-Healing of the O-RAN Open-Fronthaul Synchronization Plane · Companion to the Project Proposal
· Tanmay

A. Direct prior work — the anchor we extend (read in full)

- [1] J. Groen, S. Di Valerio, I. Karim, D. Villa, Y. Zhang, L. Bonati, M. Polese, S. D'Oro, T. Melodia, E. Bertino, F. Cuomo, K. Chowdhury, "TIMESAFE: Timing Interruption Monitoring and Security Assessment for Fronthaul Environments," **ACM Transactions on Privacy and Security**, vol. 28(5), 2025. DOI 10.1145/3775060 (arXiv:2412.13049). Code/dataset: github.com/genesys-neu/s-plane_security. — *Provides the threat model, the ~2 s base-station crash, the 97.5% detection baseline, and released attack scripts/data/twin. Our work adds the response step it leaves open.*

B. Standards & specifications — define the domain and the numbers

- [2] O-RAN ALLIANCE WG4, "Control, User and Synchronization Plane Specification (O-RAN.WG4.CUS.0)," incl. LLS-C1–C4 synchronization configurations.
- [3] O-RAN ALLIANCE WG11, "Security Threat Modeling and Remediation Analysis (Threat Model)" — lists PTP master-clock spoofing on the Open Fronthaul as a threat.
- [4] IEEE Std 1588-2019, Precision Time Protocol (PTP) for Networked Measurement and Control Systems.
- [5] ITU-T G.8275.1 (PTP telecom profile, phase/time); G.8273.2 (time-error limits for telecom boundary clocks — the ~100 ns budget); G.8262 (Synchronous Ethernet).

C. Related literature — context & positioning (★ = read in full)

- [6] ★ A. Wadud, F. Golpayegani, N. Afraz, "Conflict Management in the Near-RT-RIC of Open RAN: A Game-Theoretic Approach," IEEE iThings/GreenCom/CPSCoM, 2023.
- [7] ★ N. Varshney, C. Puligheddu, A. Badawy, C. F. Chiasserini, "XAI4C: Explainable AI for Conflict Detection and Mitigation in O-RAN Near-RT RIC," IEEE Vehicular Technology Magazine, 2025. DOI 10.1109/MVT.2025.3625693.
- [8] ★ "Conflict Mitigation Framework and Conflict Detection in O-RAN Near-RT RIC," IEEE (doc 10121578), 2023.
- [9] ★ H. Bilen et al., "KDN-Driven Zero-Shot Learning for Intelligent Self-Healing in 6G Small Cell Networks," Ad Hoc Networks 178:103984, 2025 (Elsevier).
- [10] ★ A. D. Salman et al., "Hy-LIFT: Hybrid LLM-Assisted Fault Diagnosis Framework for 5G/6G Networks," MDPI Computers 14(12):551, 2025.
- [11] ★ M. Liyanage et al., "A Survey on Zero-touch Network and Service Management (ZSM) for 5G and beyond," Elsevier J. Network & Computer Applications 203, 2022.
- [12] ★ I. Ullah et al., "Autonomous Network Management for 6G: A Comprehensive Survey," Elsevier Digital Communications & Networks 11, 2025.
- [13] "OpenTwin: Digital Twin Driven Closed-Loop KPM Inference and Control for Open RAN," arXiv:2605.24662.
- [14] L. Bonati et al., "Colosseum: The Open RAN Digital Twin," arXiv:2404.17317, 2024.
- [15] "Simba: Root-Cause Analysis of Anomalies in 5G RAN using GNN + Transformer," arXiv:2406.15638.
- [16] "How to Bridge the Sim-to-Real Gap in Digital Twin-Aided Telecommunication Networks," arXiv:2507.07067, 2025.
- [17] "KPI Poisoning: An Attack in the Open RAN Near-Real-Time Control Loop," arXiv:2505.05537.
- [18] "Towards a Multi-Layer Defence Framework for Securing Near-RT Operations in Open RAN," arXiv:2512.01596.
- [19] "ARGOS: Anomaly Recognition and Guarding through O-RAN Sensing," arXiv:2506.06916.
- [20] O-RAN nGRG, "Research Report on Digital-Twin RAN Use Cases," RS-01, 2024; and "Digital Twin for Open RAN: Towards Intelligent and Resilient 6G RANs," IEEE Communications Magazine, 2023.
- [21] "Uncertainty-Aware Digital Twins" (arXiv:2501.10337, 2025); "End-to-End O-RAN Security Architecture & the Open Fronthaul" (arXiv:2304.05513); "Failure Management in 5G RAN: Challenges and Open Research Lines" (survey).

D. Tools & software used to build the prototype

- [22] Python 3, NumPy, pandas, scikit-learn (Random Forest), Matplotlib, PyYAML, pytest — used to implement the simulator, injectors, discriminator, twin, healing loop, and benchmark.
- [23] Planned for the real-timing tier (not yet used): linuxptp (ptp4l / phc2sys / ts2phc), tc netem; srsRAN Project & FlexRIC (EURECOM) O-RAN reference stack.

Grounding: the design rests on the anchor paper [1] and the O-RAN / IEEE / ITU-T standards [2]–[5]; group C positions the work and evidences the gap. Items marked ★ were read in full; the rest are cited at abstract/summary level. Contribution type: integration + experimental validation + released dataset — not a new fundamental algorithm.